PRIMED: Joint Loss Estimation with Selective Penalization for Multi-Reader Imaging Data with Binary Outcomes

Yonghan Kwon[1] and Kyunghwa Han[1*]

[1*]Department of Radiology, Research Institute of Radiological Science, and Center for Clinical Imaging Data Science, Yonsei University College of Medicine, Seoul, Republic of Korea.

*Corresponding author(s). E-mail(s): khhan@yuhs.ac;
Contributing authors: yonghankwon0@yuhs.ac;

Abstract

Background: When multiple readers independently assess the same medical image, their interpretations frequently disagree. Conventional approaches treat this inter-reader variability as nuisance to be eliminated, yet it may carry predictive information. Existing methods for incorporating variability focus on single longitudinal biomarkers and do not address the high-dimensional, multivariate setting of multi-reader imaging data where variable selection is required.
Methods: We propose PRIMED (Penalized Regularization for Integrated Multipath Estimation with Dispersion), a joint loss framework that integrates three information channels: consensus (mean reader score), disagreement (standard deviation across readers), and the structural relationship between them. A dispersion model captures how each feature's consensus determines its disagreement, and only the residual disagreement (the component not explained by consensus) enters the outcome model. A group LASSO penalty on the outcome coefficients performs variable selection, while dispersion parameters remain unpenalized. Performance is evaluated through simulation studies and two applications: a multi-center, multi-reader MRI study of 149 rectal cancer patients (14 features, 5 readers) and the LIDC-IDRI lung nodule dataset (107 radiomics features, 1,390 nodules, 3–4 readers).

Results: In simulations across multiple settings (varying signal-to-noise ratio, noise levels, and dimensionality up to 100 features), PRIMED consistently achieves the highest F1 score and AUC, with advantages driven primarily by higher positive predictive value. Variable selection stability, measured by the Jaccard index, is highest for PRIMED at all dimensionalities (e.g., 0.228 vs. 0.182 and 0.192 at 100 features). In the rectal cancer application, PRIMED selects features missed by competing methods. In the LIDC-IDRI application, PRIMED achieves comparable prediction (10-fold cross-validated AUC 0.900) with substantially higher selection stability (Jaccard 0.798 vs. 0.688 and 0.560). **Conclusions:** PRIMED provides a principled framework for exploiting inter-reader disagreement in multi-reader imaging studies. By jointly modeling the consensus–disagreement relationship and using residual disagreement as a predictor, it achieves superior variable selection with consistent stability across replications, while maintaining competitive prediction performance. Validation on two distinct imaging modalities (MRI and CT) and scales (14 to 107 features) confirms these advantages generalize across clinical settings.

Keywords: Multi-reader imaging, Variable selection, Group LASSO, Inter-reader disagreement, Consensus–disagreement framework, Rectal cancer, Radiomics, LIDC-IDRI, Lung nodule

1 Background

When multiple readers independently assess the same medical image, their interpretations frequently disagree. This inter-reader variability is ubiquitous across imaging modalities and clinical tasks: radiologists differ in contouring tumor boundaries on MRI [1, 2], in scoring subjective features such as fibrosis or diffusion restriction, and in delineating nodule margins on CT [3]. The conventional approach treats such disagreement as nuisance to be eliminated. Reliability coefficients such as the intraclass correlation coefficient [4] and kappa [5] quantify it, summary strategies (averaging, majority voting) remove it, and measurement error methods such as regression calibration and SIMEX [6, 7] correct for the attenuation it induces. All of these strategies discard variability rather than use it.

Yet a growing body of evidence suggests that variability itself can carry predictive information. In other areas of medicine, variability has been recognized as a predictor

in its own right: visit-to-visit blood pressure variability predicts stroke and cardiovascular events independently of mean blood pressure [8, 9]. Joint models [10, 11] and mixed-effects location-scale models [12] have been developed to incorporate within-subject variability into outcome prediction. These approaches, however, focus on a single biomarker measured longitudinally and do not address the high-dimensional, multivariate setting of multi-reader imaging, where K features are assessed simultaneously and variable selection is required. This gap is particularly acute in modern radiomics, where hundreds of quantitative features are extracted from segmentation masks that differ across readers, producing a rich but unexploited source of inter-reader disagreement information.

This raises a fundamental question: when multiple readers assess the same image, should their disagreement simply be eliminated, or can it be harnessed for prediction?

A natural starting point is to treat both the consensus (mean of reader scores, $\bar{X}_k$) and the disagreement (standard deviation, S_k) as predictors in a penalized regression, a perspective we term the consensus–disagreement (CD) framework. This approach, however, faces an important obstacle: $\bar{X}_k$ and S_k are generally not independent. On bounded ordinal scales, a boundary effect forces disagreement toward zero when the consensus is near the extremes [13, 14]. Beyond this mechanical constraint, reader-specific factors such as differential confidence at different score levels produce feature-specific consensus–disagreement relationships whose direction and strength vary across features. If S_k is entered alongside $\bar{X}_k$ without accounting for this dependence, a feature can appear to carry disagreement signal simply because its disagreement tracks its consensus level, not because residual disagreement predicts the outcome. In high-dimensional settings with many inactive features, this conflation inflates false positives and destabilizes variable selection.

A straightforward implementation of the CD framework is to apply a group penalty [15] to each feature's consensus–disagreement pair, or to penalize only the consensus

mean via the LASSO [16]. Either approach treats the two summaries as independent inputs and ignores the structural relationship described above.

To address this gap, we propose PRIMED (Penalized Regularization for Integrated Multi-path Estimation with Dispersion), a joint loss framework that integrates three information channels: consensus, disagreement, and the structural relationship between them. The key idea is to fit a dispersion model that captures how each feature's consensus determines its disagreement, and to feed only the *residual* disagreement, the component not explained by consensus, into the outcome model. Because the outcome model depends on these residuals, the dispersion parameters α_k receive gradient contributions from both the dispersion and outcome losses, yielding genuinely joint estimation rather than a two-stage procedure. A group penalty is applied only to the outcome coefficients (β_k, γ_k), so that variable selection reflects outcome relevance alone. We demonstrate the framework through simulation studies and two real-data applications: a multi-reader MRI study of rectal cancer ($K = 14$, $m = 5$ readers) and a high-dimensional radiomics study of lung nodule malignancy ($K = 107$, $m = 3$–4 readers).

2 Methods

2.1 Data structure

Consider n subjects, K imaging features, and m readers. Let x_{ijk} denote the score assigned by reader j ($j = 1, \ldots, m$) to subject i ($i = 1, \ldots, n$) on feature k ($k = 1, \ldots, K$). For each subject–feature pair, define the consensus (sample mean across readers) and the disagreement (sample standard deviation across readers); alternative choices of these summary statistics for different measurement scales are discussed in

Appendix A:

$$\bar{X}_{ik} = \frac{1}{m}\sum_{j=1}^{m} x_{ijk}, \qquad S_{ik} = \left[\frac{1}{m-1}\sum_{j=1}^{m}(x_{ijk} - \bar{X}_{ik})^2\right]^{1/2}.$$

If the number of readers varies across subjects (e.g., due to missing evaluations or, as in the LIDC-IDRI application, varying numbers of annotators per nodule), m is replaced by the subject-specific count m_i in the definitions above. Let $Y_i \in \{0, 1\}$ denote the binary outcome for subject i. The observed data thus consist of the consensus matrix $\bar{X} \in \mathbb{R}^{n\times K}$, the disagreement matrix $S \in \mathbb{R}^{n\times K}$, and the outcome vector $Y \in \{0, 1\}^n$.

2.2 Consensus–disagreement framework

Rather than discarding S, we treat both $\bar{X}$ and S as potential predictors of Y, a perspective we call the consensus–disagreement (CD) framework. Two baseline strategies arise naturally within this framework. The first, which we term the *CD group LASSO* (CD GL), regresses Y on the full set $(\bar{X}_1, S_1, \ldots, \bar{X}_K, S_K)$ with a group LASSO penalty [15] that links each $(\bar{X}_k, S_k)$ pair, so that consensus and disagreement for a given feature are selected or dropped together. The second, the *consensus LASSO* (C LASSO), regresses Y on $(\bar{X}_1, \ldots, \bar{X}_K)$ with an ℓ_1 penalty [16], using only the consensus and ignoring the disagreement channel entirely.

Both methods enter $\bar{X}_k$ and S_k as independent inputs. As noted in Section 1, however, $\bar{X}_k$ and S_k are generally correlated, and this structural relationship varies in direction and strength across features. Including S_k alongside $\bar{X}_k$ without accounting for their dependence conflates the direct contribution of disagreement with the part that merely reflects the consensus level, potentially distorting coefficient estimates and variable selection. PRIMED addresses this limitation by explicitly modeling the consensus–disagreement relationship and decomposing disagreement into a structurally explained component and a residual, as described next.

2.3 Joint loss function

The PRIMED loss comprises two components (a dispersion loss and an outcome loss), each normalized by its respective null baseline. Figure 2 illustrates the overall structure: the dispersion model (top) links consensus to disagreement, and the outcome model (bottom) uses both consensus and residual disagreement as predictors.

Because the consensus and disagreement of a feature are generally correlated, entering both into an outcome model without accounting for this dependence confounds their individual contributions. The dispersion loss captures this relationship explicitly: for each feature k, a linear regression links consensus to disagreement: $S_{ik} = \alpha_{0k} + \alpha_k \bar{X}_{ik} + \varepsilon_{ik}$, where α_{0k} is the feature-specific intercept, α_k is the slope quantifying how the consensus level affects disagreement, and ε_{ik} is the residual error. The squared-error loss and its null baseline (intercept-only model, i.e., $\alpha_k = 0$ for all k) are

$$\mathcal{L}_{\text{disp}} = \frac{1}{2n} \sum_{k=1}^{K} \sum_{i=1}^{n} (S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik})^2, \qquad \mathcal{L}^0_{\text{disp}} = \frac{1}{2} \sum_{k=1}^{K} \widehat{\text{Var}}(S_k).$$

The dispersion model defines, for each subject–feature pair, the residual disagreement

$$\tilde{S}_{ik} = S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik},$$

i.e., the part of reader disagreement that is not explained by the consensus level. This residual is the quantity used in the outcome model, ensuring that γ_k captures the effect of disagreement *beyond* what is structurally linked to the consensus.

The outcome model is a logistic regression of Y_i on both the consensus and the residual disagreement predictors. Let β_0 denote the intercept, β_k the coefficient for the consensus predictor of feature k, and γ_k the coefficient for the residual disagreement predictor of feature k. The predicted probability for subject i is $p_i =$

$\text{logit}^{-1}\big(\beta_0 + \sum_{k=1}^{K} \beta_k \bar{X}_{ik} + \sum_{k=1}^{K} \gamma_k \tilde{S}_{ik}\big)$. If α_k were estimated by OLS on the dispersion model alone, $\tilde{S}_k$ would be exactly sample-orthogonal to $\bar{X}_k$ by construction, and β_k and γ_k would capture non-overlapping components of the linear predictor. In the joint estimation, the coupling gradient (Section 2.5) can shift α_k away from the OLS solution when $\gamma_k \neq 0$, so the exact orthogonality is approximate rather than exact; it holds precisely for features with $\gamma_k = 0$.

The negative log-likelihood and its null baseline (intercept-only model, predicting the marginal proportion $\bar{Y} = n^{-1} \sum_i Y_i$) are

$$\mathcal{L}_{\text{out}} = -\frac{1}{n} \sum_{i=1}^{n} \big[Y_i \log p_i + (1-Y_i) \log(1-p_i)\big], \qquad \mathcal{L}^0_{\text{out}} = -\big[\bar{Y} \log \bar{Y} + (1-\bar{Y}) \log(1-\bar{Y})\big].$$

Because the outcome loss depends on the residual disagreement $\tilde{S}_{ik}$, which is itself a function of α, the joint loss couples α to both components:

$$\mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = \frac{\mathcal{L}_{\text{disp}}(\boldsymbol{\alpha})}{\mathcal{L}^0_{\text{disp}}} + \frac{\mathcal{L}_{\text{out}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma})}{\mathcal{L}^0_{\text{out}}}. \tag{1}$$

Each component is normalized by its null value so that both terms start at 1.0 under the null model, eliminating the need for a weighting hyperparameter. The dependence of $\mathcal{L}_{\text{out}}$ on $\boldsymbol{\alpha}$ through $\tilde{S}_{ik}$ is the mechanism that achieves genuinely joint estimation of the dispersion and outcome parameters.

2.4 Penalty structure

Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)^T$, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)^T$, and $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_K)^T$ collect the slope coefficients across all features, and let $\lambda > 0$ denote the regularization parameter controlling sparsity. The penalized objective is

$$\min_{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}} \; \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) \; + \; \lambda \sum_{k=1}^{K} \sqrt{\beta_k^2 + \gamma_k^2}. \tag{2}$$

The penalty is a group LASSO norm applied to the outcome coefficients (β_k, γ_k) for each feature k. The group structure ensures that each feature is evaluated holistically (considering both its consensus signal (β_k) and its residual disagreement signal (γ_k)), rather than allowing one channel to be selected in isolation, which could lead to the inclusion of features with a spurious consensus or disagreement effect. Since each group consistently contains two coefficients, group-specific weights (such as $\sqrt{p_g}$ scaling) are unnecessary and are omitted.

The intercepts α_{0k} and β_0 are unpenalized, as is standard. Crucially, the dispersion slopes α_k are also unpenalized: they are updated by gradient descent without penalty-induced shrinkage. Although α_k receives no penalty, it is *not* estimated in a separate stage: because $\mathcal{L}_{\text{out}}$ depends on α_k through $\tilde{S}_{ik}$, the gradient of the outcome loss with respect to α_k provides an additional update signal (see Section 2.5). This selective penalization ensures that the consensus–disagreement relationship is estimated without penalty-induced bias regardless of outcome relevance, while variable selection is driven solely by the outcome coefficients (β_k, γ_k).

All predictors ($\bar{X}_k$ and S_k) are standardized to zero mean and unit variance within the training set prior to penalization, ensuring that the group LASSO penalty is applied on a common scale across features with different units.

Feature k is selected if $\hat{\beta}_k \neq 0$ or $\hat{\gamma}_k \neq 0$.

2.5 Optimization

We minimize (2) via proximal gradient descent (Algorithm 1). Let $\eta > 0$ denote the step size (learning rate). At each iteration, all parameters receive a gradient step. The dispersion parameters (α_{0k}, α_k) receive gradients from *two* sources: the dispersion loss $\mathcal{L}_{\text{disp}}$ and, because $\tilde{S}_{ik} = S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik}$ enters the outcome model, the outcome

loss $\mathcal{L}_{\text{out}}$. The coupling gradient for α_k from the outcome loss is

$$\frac{\partial \mathcal{L}_{\text{out}}}{\partial \alpha_k} = \frac{\gamma_k}{n\,\mathcal{L}^0_{\text{out}}} \sum_{i=1}^{n} (p_i - Y_i)\,\bar{X}_{ik},$$

which arises because $\partial \tilde{S}_{ik}/\partial \alpha_k = -\bar{X}_{ik}$. This gradient vanishes when $\gamma_k = 0$ (i.e., when feature k is not selected), so only outcome-relevant features couple the two losses. More precisely, at any stationary point the gradient of $\mathcal{L}$ with respect to α_k satisfies

$$\underbrace{\frac{1}{n\,\mathcal{L}^0_{\text{disp}}} \sum_{i=1}^{n} \tilde{S}_{ik}\,\bar{X}_{ik}}_{\text{dispersion gradient}} \;=\; \underbrace{\frac{\gamma_k}{n\,\mathcal{L}^0_{\text{out}}} \sum_{i=1}^{n} (p_i - Y_i)\,\bar{X}_{ik}}_{\text{coupling gradient}}. \tag{3}$$

Because the joint objective is non-convex (see below), (3) characterizes any stationary point reached by the algorithm, not a unique global solution. At such a point, when $\gamma_k = 0$ the right-hand side vanishes, so $\sum_i \tilde{S}_{ik}\,\bar{X}_{ik} = 0$: the residual $\tilde{S}_k$ is sample-orthogonal to $\bar{X}_k$, and $\hat{\alpha}_k$ coincides with the ordinary least squares estimate. When $\gamma_k \neq 0$, $\hat{\alpha}_k$ departs from the OLS solution; the coupling gradient shifts α_k in a direction that reduces the joint loss, producing a purposeful bias in the dispersion model that benefits outcome prediction. The outcome intercept β_0 is updated by standard gradient descent. The outcome coefficients (β_k, γ_k) then receive a group-wise proximal (soft-thresholding) step:

$$(\beta_k, \gamma_k) \leftarrow \begin{cases} \Big(1 - \dfrac{\eta\lambda}{\|(\tilde{\beta}_k, \tilde{\gamma}_k)\|_2}\Big)(\tilde{\beta}_k, \tilde{\gamma}_k), & \text{if } \|(\tilde{\beta}_k, \tilde{\gamma}_k)\|_2 > \eta\lambda, \\ (0, 0), & \text{otherwise,} \end{cases}$$

where $(\tilde{\beta}_k, \tilde{\gamma}_k)$ denotes the post-gradient values. This proximal operator is the key mechanism that sets entire (β_k, γ_k) pairs exactly to zero, producing group-wise

sparsity. The dispersion coefficients α_k are never shrunk, ensuring that the consensus–disagreement relationship is estimated without penalty-induced bias regardless of outcome relevance.

Convergence is declared when the relative change in parameters falls below a tolerance $\epsilon = 10^{-6}$, or after a maximum of 10,000 iterations. In practice, the step size $\eta = 0.005$ provides stable convergence across all settings examined.

We note that the joint objective (2) is non-convex in $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma})$ jointly, because $\mathcal{L}_{\text{out}}$ contains the bilinear product $\gamma_k \alpha_k$ through the residual $\tilde{S}_{ik}$. The objective is convex in $\boldsymbol{\alpha}$ with $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ fixed, and convex in $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ with $\boldsymbol{\alpha}$ fixed, so the proximal gradient algorithm converges to a stationary point but not necessarily a global minimum. In our experiments, convergence was robust to initialization and we did not observe sensitivity to local minima.

A practical concern with non-convex objectives is that the regularization path may be non-monotone: decreasing λ could, in principle, cause previously selected features to drop out, because the change in $\hat{\alpha}_k$ alters $\tilde{S}_{ik}$ and thereby reshapes the loss landscape. In convex group LASSO, path monotonicity (the number of selected features increases as λ decreases) is mathematically guaranteed; no such guarantee exists for PRIMED. To assess this empirically, we traced the full regularization path on the LIDC-IDRI dataset ($K = 107$), evaluating 50 log-spaced λ values from $\lambda_{\max}$ (0 features selected) to $\lambda_{\max}/1000$ (all 107 features selected), each from a cold start (zero initialization). The path was strictly monotone: the number of selected features increased or remained constant at every step, with zero violations across 49 transitions. Figure 1 visualizes the result: the number of selected features increases smoothly from 0 (at $\lambda_{\max}$) to 107 (at $\lambda_{\max}/1000$), with no reversals. This suggests that, despite the theoretical non-convexity, the bilinear coupling between α_k and γ_k does not disrupt path regularity in practice, likely because the dispersion model is well identified for

most features and the coupling gradient produces only small perturbations around the OLS solution for α_k.

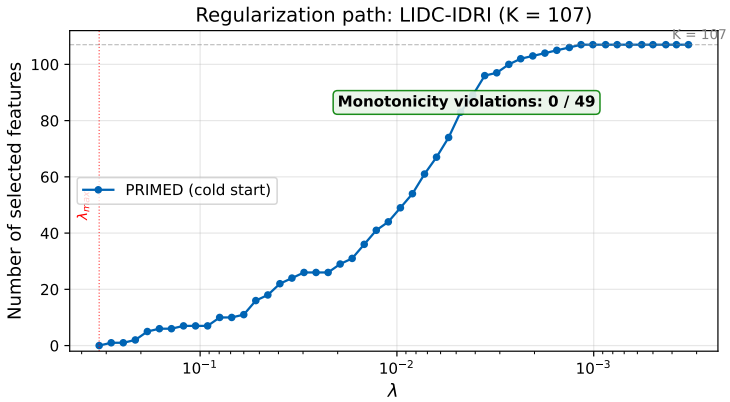

Fig. 1 Regularization path for PRIMED on the LIDC-IDRI dataset. Each point represents an independent fit from zero initialization (cold start) at a given λ with $K = 107$ features. The number of selected features increases monotonically as λ decreases, with zero violations across 49 transitions. Despite the non-convexity of the joint objective, the path exhibits the same monotone behavior as convex group LASSO.

2.6 Coefficient interpretation

Substituting $\tilde{S}_{ik} = S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik}$ into the linear predictor η_i yields the algebraic identity

$$\eta_i = \underbrace{\left(\beta_0 - \textstyle\sum_k \gamma_k \alpha_{0k}\right)}_{\text{intercept}} + \sum_{k=1}^{K} \underbrace{\left(\beta_k - \gamma_k \alpha_k\right)}_{\beta_k^{\mathrm{CD}}} \bar{X}_{ik} + \sum_{k=1}^{K} \gamma_k S_{ik},$$

where $\beta_k^{\mathrm{CD}} = \beta_k - \gamma_k \alpha_k$ is the coefficient that $\bar{X}_k$ would receive if S_k (rather than $\tilde{S}_k$) were entered directly as a predictor, as in CD GL. This identity holds regardless of the link function, because it concerns only the structure of the linear predictor. Rearranging gives

$$\beta_k = \beta_k^{\mathrm{CD}} + \gamma_k \alpha_k, \tag{4}$$

which decomposes the association of $\bar{X}_k$ with Y (on the linear predictor scale) into a direct component β_k and an indirect component $\gamma_k \alpha_k$ mediated through disagreement:

$$\underbrace{\beta_k^{\text{total}}}_{\text{total}} = \underbrace{\beta_k}_{\text{direct}} + \underbrace{\alpha_k \cdot \gamma_k}_{\text{indirect via disagreement}} . \tag{5}$$

This decomposition is structurally similar in form to the total/direct/indirect effect partition used in mediation analysis, but it is applied here to the linear predictor of a penalized regression and does not carry a causal interpretation. It serves as a bookkeeping device that clarifies how PRIMED's coefficient β_k relates to the CD GL coefficient β_k^{CD} and quantifies the extent to which the consensus–disagreement relationship (α_k) mediates the association between $\bar{X}_k$ and Y through the disagreement channel.

When $|\hat{\alpha}_k|$ is large and $r(\bar{X}_k, S_k)$ is close to ± 1, the residual $\tilde{S}_k$ contains little variance, which naturally limits the precision of $\hat{\gamma}_k$. This is not a deficiency of the method but rather an informative outcome: a near-deterministic consensus–disagreement relationship leaves little residual disagreement available to predict the outcome, and PRIMED correctly reflects this by yielding a small or zero $\hat{\gamma}_k$. The group selection decision remains unaffected, as it depends on the joint norm $\|(\hat{\beta}_k, \hat{\gamma}_k)\|_2$ rather than on the individual coefficients.

2.7 Regularization path and cross-validation

The regularization parameter λ controls the sparsity of the solution: larger λ sets more (β_k, γ_k) pairs to zero. We construct a data-driven regularization path, analogous to the approach used in `glmnet` [17]. Let $\lambda_{\max}$ denote the smallest λ for which all $(\beta_k, \gamma_k) = (0, 0)$, computed as the maximum group gradient norm at the intercept-only solution (see Appendix B for derivation). A log-spaced grid of $L = 100$ values from $\lambda_{\max} \cdot \epsilon$ to $\lambda_{\max}$ adapts automatically to the scale of the data, avoiding sensitivity

to a fixed grid specification. We use $\epsilon = 0.01$ in the simulations and $\epsilon = 0.001$ in the LIDC-IDRI application, where the larger feature space benefits from finer resolution at small λ (see Appendix B).

The optimal λ is selected by stratified five-fold cross-validation. The data are split into five folds, stratifying by Y to maintain class balance. For each candidate λ_j and fold f:

1. Fit the PRIMED model on the training set (folds $\neq f$).
2. Compute the residual disagreement for the held-out fold using the training-set estimates $\hat{\alpha}_{0k}$ and $\hat{\alpha}_k$: $\tilde{S}_{ik} = S_{ik} - \hat{\alpha}_{0k} - \hat{\alpha}_k \bar{X}_{ik}$ for $i \in$ fold f.
3. Evaluate the outcome deviance on the held-out fold f:

$$D_f(\lambda_j) = -\frac{2}{n_f} \sum_{i \in \text{fold } f} \left[Y_i \log \hat{p}_i + (1 - Y_i) \log(1 - \hat{p}_i) \right].$$

The cross-validated deviance is $\mathrm{CV}(\lambda_j) = \frac{1}{5} \sum_{f=1}^{5} D_f(\lambda_j)$. The selected $\lambda^* = \arg\min_j \mathrm{CV}(\lambda_j)$, and the final model is refit on the full dataset with λ^*.

Only the outcome deviance (not the dispersion loss) is used for λ selection, because the penalty applies exclusively to (β_k, γ_k).

2.8 Nonlinear dispersion model

The linear dispersion model $S_{ik} = \alpha_{0k} + \alpha_k \bar{X}_{ik} + \varepsilon_{ik}$ is the simplest choice, but the consensus–disagreement relationship need not be linear. On bounded ordinal scales, a boundary effect forces disagreement toward zero when the consensus is near either extreme, producing an inverted-U-shaped relationship that a linear model cannot capture [13, 14]. PRIMED accommodates nonlinear dispersion by replacing the linear model with a polynomial. In particular, a quadratic dispersion model

$$S_{ik} = \alpha_{0k} + \alpha_{1k} \bar{X}_{ik} + \alpha_{2k} \bar{X}_{ik}^2 + \varepsilon_{ik} \tag{6}$$

captures the inverted-U pattern with a single additional parameter per feature. The residual disagreement becomes $\tilde{S}_{ik} = S_{ik} - \alpha_{0k} - \alpha_{1k}\bar{X}_{ik} - \alpha_{2k}\bar{X}_{ik}^2$, and the coupling gradients from the outcome loss generalize naturally:

$$\frac{\partial \mathcal{L}_{\text{out}}}{\partial \alpha_{1k}} = \frac{\gamma_k}{n\,\mathcal{L}_{\text{out}}^0} \sum_{i=1}^{n} (p_i - Y_i)\,\bar{X}_{ik}, \qquad \frac{\partial \mathcal{L}_{\text{out}}}{\partial \alpha_{2k}} = \frac{\gamma_k}{n\,\mathcal{L}_{\text{out}}^0} \sum_{i=1}^{n} (p_i - Y_i)\,\bar{X}_{ik}^2.$$

All other components of the framework (the outcome model, the group penalty on (β_k, γ_k), and the proximal gradient algorithm) remain unchanged. Both α_{1k} and α_{2k} are unpenalized, consistent with the selective penalization principle. In Section 3.4 we compare the linear and quadratic dispersion models on the rectal cancer data.

2.9 Comparison methods

We compare PRIMED against the two baseline strategies introduced in Section 2.2. The CD group LASSO (CD GL) is implemented via `grpreg`, and the consensus LASSO (C LASSO) via `glmnet`. For both methods, λ is selected by stratified five-fold cross-validation using the same fold assignments as PRIMED to ensure a fair comparison.

Both methods can be viewed as limiting cases of the PRIMED objective. If α_k is constrained to zero for all k, the normalized dispersion loss $\mathcal{L}_{\text{disp}}/\mathcal{L}_{\text{disp}}^0$ becomes constant at 1.0 and $\tilde{S}_{ik}$ reduces to $S_{ik} - \bar{S}_k$; the outcome loss then becomes a group LASSO on $(\bar{X}_k, S_k - \bar{S}_k)$, which has the same functional form as CD GL but uses centered disagreement rather than raw S_k. If γ_k is constrained to zero for all k, the group penalty reduces to $\lambda|\beta_k|$, and the outcome model depends only on the consensus, recovering C LASSO. These correspondences are structural: the actual coefficient estimates may differ because PRIMED and the baseline methods use different optimization procedures and loss functions.

3 Results

3.1 Noise variable simulation

For each subject i and feature k, the data are generated as:

$$
\begin{aligned}
\mu_{ik} &\sim N(0,1), \\
\sigma_{ik} &= \exp(\alpha_k \mu_{ik} + \varepsilon_{ik}), \quad \varepsilon_{ik} \sim N(0, \tau^2), \\
x_{ijk} &\sim N(\mu_{ik}, \sigma_{ik}^2), \quad j = 1, \ldots, m, \\
P(Y_i = 1) &= \text{logit}^{-1}\Big(\beta_0 + \textstyle\sum_k \beta_k \mu_{ik} + \sum_k \gamma_k \sigma_{ik}\Big).
\end{aligned}
\tag{7}
$$

The parameter τ controls the noise in the dispersion channel: as τ increases, S_k becomes a noisier proxy for the latent σ_{ik}.

Following standard variable selection terminology [15, 16], we classify the K features into three types based on their role in the data generating process:

- *Active* features ($\beta_k \neq 0$ or $\gamma_k \neq 0$): outcome-relevant features whose consensus or disagreement contributes to Y.
- *Noise* features ($\alpha_k \neq 0$, $\beta_k = \gamma_k = 0$): features exhibiting a consensus–disagreement relationship (reader effect) but no outcome relevance. These are the most challenging for variable selection, as their nontrivial dispersion structure may be mistaken for an outcome signal.
- *Null* features ($\alpha_k = \beta_k = \gamma_k = 0$): features with no effect on any path.

Two data generating processes are considered:

- **DGP 1** ($\gamma_k \neq 0$): Disagreement directly contributes to the outcome through γ_k. For active features, $\alpha_k = \beta_k = \gamma_k = c$ (set equal for simplicity, so that all three channels contribute at the same magnitude).

- **DGP 2** ($\gamma_k = 0$): Disagreement exists (via $\alpha_k \neq 0$) but does not affect the outcome. For active features, $\alpha_k = \beta_k = c$ and $\gamma_k = 0$.

DGP 1 tests whether PRIMED can exploit the disagreement signal, while DGP 2 tests whether it avoids false selection when disagreement is outcome-irrelevant.

$K = 20$ features: 7 active, 3 noise ($\alpha_k = c$, $\beta_k = \gamma_k = 0$), and 10 null ($\alpha_k = \beta_k = \gamma_k = 0$). Sample size $n = 200$, replications $B = 50$, replicates $m = 5$, $\tau = 0.3$. Figure 3 illustrates the variable structure.

All active features share a common coefficient magnitude c. We set c so that the McKelvey–Zavoina signal-to-noise ratio $\mathrm{SNR} = \mathrm{Var}(\eta)/(\pi^2/3)$ equals a target value, where $\eta = \beta_0 + \sum_k \beta_k \mu_{ik} + \sum_k \gamma_k \sigma_{ik}$ is the linear predictor. Two settings are examined: SNR = 0.5 and SNR = 1.0. The intercept β_0 is calibrated so that $E[Y] \approx 0.5$.

Two scenarios are compared:

- **S1**: No noise variables ($\alpha_{8\text{–}10} = 0$).
- **S2**: Noise present ($\alpha_{8\text{–}10} = c$, $\beta_{8\text{–}10} = \gamma_{8\text{–}10} = 0$).

Variable selection performance is evaluated by comparing the set of features selected by each method against the true active set. TPR (true positive rate) is the proportion of truly active features that are selected, PPV (positive predictive value) is the proportion of selected features that are truly active, and F1 is their harmonic mean: $\mathrm{F1} = 2 \cdot \mathrm{PPV} \cdot \mathrm{TPR}/(\mathrm{PPV} + \mathrm{TPR})$. Prediction performance is measured by the area under the receiver operating characteristic curve (AUC) on an independently generated test set from the same DGP. All metrics are averaged over B replications.

Table 1 reports F1 and AUC for DGP 1. PRIMED achieves the highest AUC at both SNR levels under both S1 and S2, and the highest F1 in most settings. At SNR = 1.0 (S1), PRIMED achieves F1 of .730 versus .702 (CD GL) and .699 (C LASSO).

Table 2 reports results for DGP 2, when disagreement does not contribute to the outcome. PRIMED again achieves the highest F1 and AUC across all settings.

At SNR = 1.0 (S2), PRIMED achieves F1 of .740 versus .704 (CD GL) and .705 (C LASSO), with the difference driven primarily by higher PPV.

Table 1 Noise variable simulation: DGP 1 ($\gamma \neq 0$, K=20, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.909** (*.175*)	.590 (*.151*)	**.703** (*.130*)	**.695** (*.040*)	**.989** (*.049*)	**.592** (*.140*)	**.730** (*.101*)	**.764** (*.024*)
	CD GL	.894 (*.128*)	**.598** (*.149*)	.700 (*.090*)	.684 (*.029*)	.971 (*.091*)	.560 (*.111*)	.702 (*.093*)	.750 (*.029*)
	C LASSO	.909 (*.115*)	.571 (*.147*)	.685 (*.099*)	.688 (*.025*)	.974 (*.055*)	.553 (*.110*)	.699 (*.087*)	.742 (*.023*)
S2	PRIMED	.909 (*.173*)	.584 (*.146*)	.699 (*.129*)	**.695** (*.040*)	**.983** (*.069*)	**.580** (*.146*)	**.717** (*.100*)	**.763** (*.025*)
	CD GL	.891 (*.128*)	**.608** (*.160*)	**.705** (*.098*)	.683 (*.029*)	.971 (*.091*)	.559 (*.110*)	.701 (*.090*)	.750 (*.028*)
	C LASSO	**.914** (*.112*)	.571 (*.141*)	.688 (*.096*)	.688 (*.024*)	.974 (*.063*)	.558 (*.110*)	.703 (*.087*)	.742 (*.023*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

Table 2 Noise variable simulation: DGP 2 ($\gamma = 0$, K=20, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.926** (*.120*)	**.635** (*.138*)	**.739** (*.095*)	**.699** (*.026*)	**.986** (*.043*)	**.624** (*.121*)	**.758** (*.091*)	**.757** (*.020*)
	CD GL	.889 (*.164*)	.623 (*.171*)	.704 (*.111*)	.679 (*.028*)	.977 (*.060*)	.560 (*.120*)	.705 (*.094*)	.740 (*.025*)
	C LASSO	.914 (*.158*)	.579 (*.151*)	.699 (*.136*)	.692 (*.036*)	.966 (*.074*)	.566 (*.118*)	.705 (*.088*)	.734 (*.022*)
S2	PRIMED	**.923** (*.123*)	.617 (*.131*)	**.727** (*.098*)	**.698** (*.026*)	**.977** (*.073*)	**.608** (*.126*)	**.740** (*.093*)	**.754** (*.022*)
	CD GL	.874 (*.177*)	**.635** (*.180*)	.704 (*.114*)	.677 (*.029*)	.977 (*.053*)	.558 (*.114*)	.704 (*.088*)	.739 (*.024*)
	C LASSO	.914 (*.158*)	.576 (*.151*)	.697 (*.136*)	.693 (*.036*)	.966 (*.074*)	.565 (*.104*)	.705 (*.079*)	.735 (*.021*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

3.2 Sensitivity to subject-specific noise (τ)

Using the same $K = 20$ structure as Section 3.1, we vary $\tau \in \{0.3, 0.6, 0.9\}$ under all four DGP×scenario combinations. As τ increases, subject-specific noise ε_{ik} dominates the dispersion channel: at $\tau = 0.9$, $\mathrm{Var}(\sigma_k)$ is roughly 11 times its value at $\tau = 0.3$, making S_k a much noisier proxy for σ_{ik}. The coefficient c is recalibrated at each (τ, SNR) combination to hold SNR constant, so that any change in performance is due to τ rather than signal strength.

Table 3 reports F1 for DGP 1 (S2: noise present). C LASSO's F1 degrades sharply as τ increases: at SNR $= 0.5$, F1 drops from .688 ($\tau = 0.3$) to .286 ($\tau = 0.9$). PRIMED maintains competitive or highest F1 across τ levels (Figure 4). At SNR $=$ 1.0, PRIMED achieves F1 of .717/.712/.700 versus CD GL's .701/.681/.683.

Table 4 extends this analysis to all four DGP×scenario combinations. Under DGP 1 (S1), the pattern mirrors S2 with slightly higher absolute F1 values. Under DGP 2 ($\gamma = 0$), PRIMED maintains the highest F1 at $\tau \leq 0.6$; at $\tau = 0.9$ with SNR $= 0.5$, all methods perform similarly (F1 $\approx$ .55–.59), reflecting the inherent difficulty of the regime. Notably, C LASSO does not collapse under DGP 2 as it does under DGP 1, because the outcome signal resides entirely in the consensus channel when $\gamma = 0$.

Table 3 Sensitivity to τ: DGP 1 ($\gamma \neq 0$, K=20, S2)

		TPR			PPV			F1			AUC		
		τ=.3	τ=.6	τ=.9	τ=.3	τ=.6	τ=.9	τ=.3	τ=.6	τ=.9	τ=.3	τ=.6	τ=.9
SNR=0.5	PRIMED	.909	.860	.800	.584	**.592**	.562	.699	.679	.628	**.695**	**.678**	.652
	CD GL	.891	**.874**	**.849**	**.608**	.591	**.573**	**.705**	**.686**	**.668**	.683	.675	**.653**
	C LASSO	**.914**	.694	.286	.571	.537	.353	.688	.579	.286	.688	.623	.537
SNR=1.0	PRIMED	**.983**	.969	.963	**.580**	.577	**.565**	**.717**	**.712**	**.700**	**.763**	**.752**	**.733**
	CD GL	.971	**.980**	**.966**	.559	.530	.540	.701	.681	.683	.750	.747	.726
	C LASSO	.974	.866	.443	.558	**.582**	.452	.703	.682	.405	.742	.675	.569

Mean over $B = 50$ replications; best in bold. SD values available in Table 4.

3.3 Increasing dimensionality ($K = 50, 100$)

We increase the number of features to $K = 50$ and $K = 100$, with $n = 200$, under both DGP 1 and DGP 2. All other settings match Section 3.1: $\tau = 0.3$, $m = 5$, $B = 50$. The number of active and noise features is held fixed at 7 and 3, respectively; additional features beyond $K = 20$ are null ($\alpha_k = \beta_k = \gamma_k = 0$). The coefficient c and intercept β_0 are recalibrated for each SNR.

Table 4 Sensitivity to τ: all DGP×scenario combinations (K=20)

	SNR = 0.5			SNR = 1.0		
	τ=.3	τ=.6	τ=.9	τ=.3	τ=.6	τ=.9
DGP 1 ($\gamma \neq 0$), S1						
PRIMED	**.703** (*.130*)	**.704** (*.103*)	.631 (*.194*)	**.730** (*.101*)	**.714** (*.086*)	**.711** (*.116*)
CD GL	.700 (*.090*)	.688 (*.087*)	**.670** (*.176*)	.702 (*.093*)	.689 (*.087*)	.685 (*.092*)
C LASSO	.685 (*.099*)	.585 (*.220*)	.285 (*.273*)	.699 (*.087*)	.671 (*.174*)	.420 (*.256*)
DGP 1 ($\gamma \neq 0$), S2						
PRIMED	.699 (*.129*)	.679 (*.136*)	.628 (*.195*)	**.717** (*.100*)	**.712** (*.091*)	**.700** (*.110*)
CD GL	**.705** (*.098*)	**.686** (*.088*)	**.668** (*.176*)	.701 (*.090*)	.681 (*.085*)	.683 (*.091*)
C LASSO	.688 (*.096*)	.579 (*.234*)	.286 (*.268*)	.703 (*.087*)	.682 (*.168*)	.405 (*.259*)
DGP 2 ($\gamma = 0$), S1						
PRIMED	**.739** (*.095*)	**.722** (*.127*)	**.620** (*.173*)	**.758** (*.091*)	**.763** (*.098*)	**.694** (*.123*)
CD GL	.704 (*.111*)	.655 (*.124*)	.527 (*.219*)	.705 (*.094*)	.678 (*.165*)	.563 (*.241*)
C LASSO	.699 (*.136*)	.649 (*.202*)	.546 (*.267*)	.705 (*.088*)	.670 (*.146*)	.581 (*.209*)
DGP 2 ($\gamma = 0$), S2						
PRIMED	**.727** (*.098*)	**.692** (*.136*)	**.588** (*.191*)	**.740** (*.093*)	**.753** (*.091*)	**.665** (*.145*)
CD GL	.704 (*.114*)	.632 (*.172*)	.550 (*.225*)	.704 (*.088*)	.659 (*.198*)	.560 (*.241*)
C LASSO	.697 (*.136*)	.669 (*.179*)	.558 (*.254*)	.705 (*.079*)	.672 (*.143*)	.584 (*.209*)

F1 scores (*SD*); best in bold. DGP 1: $\alpha_k = \beta_k = \gamma_k = c$; DGP 2: $\alpha_k = \beta_k = c$, $\gamma_k = 0$. S1: no noise; S2: noise present. All values averaged over $B = 50$ replications.

Tables 6 and 7 report F1 and AUC for DGP 1. PRIMED maintains a consistent AUC advantage across dimensionalities, though its F1 advantage is not strictly monotonic across K. At SNR = 1.0 (S2), PRIMED achieves F1 of .585 ($K = 50$) and .500 ($K = 100$), compared with .619/.477 (CD GL) and .546/.497 (C LASSO). Figure 5 visualizes the F1 trend across $K \in \{20, 50, 100\}$.

Tables 8 and 9 report the corresponding results for DGP 2 ($\gamma = 0$). The pattern is consistent: PRIMED achieves the highest F1 across all settings. At $K = 100$, SNR = 1.0 (S2), PRIMED achieves F1 of .591 versus .481 (CD GL) and .482 (C LASSO), a gap exceeding .100, consistent with the pattern under DGP 1. The advantage is driven by PRIMED's higher PPV, reflecting its ability to separate the dispersion structure from outcome relevance even when $\gamma = 0$.

In addition to F1 and AUC, we assess variable selection stability via the Jaccard index, averaged over all $\binom{B}{2}$ pairs of replications. Table 5 reports the results for the S2 scenario (noise present) under DGP 1 at SNR = 1.0. PRIMED achieves the highest

Jaccard index at all dimensionalities: 0.507 ($K = 20$), 0.352 ($K = 50$), and 0.228 ($K = 100$), compared to 0.485/0.306/0.182 for CD GL and 0.416/0.318/0.192 for C LASSO. All methods show declining stability with increasing K, as expected when the number of candidate features grows, but PRIMED's advantage amplifies in higher dimensions: the gap over CD GL increases from +0.022 at $K = 20$ to +0.046 at $K = 100$.

Table 5 Variable selection stability (Jaccard index) across dimensionality

Method	$K = 20$	$K = 50$	$K = 100$
PRIMED	**0.507**	**0.352**	**0.228**
CD GL	0.485	0.306	0.182
C LASSO	0.416	0.318	0.192

DGP 1 ($\gamma \neq 0$), S2, SNR = 1.0. Jaccard index averaged over all $\binom{50}{2} = 1{,}225$ pairs of $B = 50$ replications.

Table 6 Mid-dimensional simulation: DGP 1 ($\gamma \neq 0$, K=50, n=200, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.871** (*.198*)	.474 (*.168*)	**.579** (*.129*)	**.677** (*.036*)	**.954** (*.134*)	.443 (*.150*)	.580 (*.125*)	**.747** (*.034*)
	CD GL	.780 (*.249*)	**.501** (*.203*)	.564 (*.150*)	.654 (*.039*)	.937 (*.120*)	**.478** (*.157*)	**.612** (*.120*)	.733 (*.031*)
	C LASSO	.826 (*.222*)	.420 (*.163*)	.525 (*.135*)	.665 (*.041*)	.906 (*.188*)	.448 (*.184*)	.556 (*.120*)	.720 (*.036*)
S2	PRIMED	**.871** (*.198*)	.471 (*.167*)	**.576** (*.127*)	**.678** (*.036*)	**.954** (*.134*)	.450 (*.160*)	.585 (*.133*)	**.747** (*.034*)
	CD GL	.774 (*.263*)	**.494** (*.222*)	.559 (*.173*)	.650 (*.045*)	.934 (*.109*)	**.482** (*.144*)	**.619** (*.114*)	.734 (*.031*)
	C LASSO	.823 (*.221*)	.411 (*.163*)	.516 (*.133*)	.665 (*.041*)	.886 (*.225*)	.444 (*.196*)	.546 (*.143*)	.717 (*.047*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

Table 7 High-dimensional simulation: DGP 1 ($\gamma \neq 0$, K=100, n=200, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.646** (*.319*)	.418 (*.256*)	.430 (*.181*)	**.636** (*.057*)	**.909** (*.135*)	.369 (*.149*)	**.503** (*.124*)	**.727** (*.035*)
	CD GL	.586 (*.296*)	.368 (*.228*)	.401 (*.186*)	.615 (*.056*)	.869 (*.175*)	.356 (*.156*)	.469 (*.112*)	.700 (*.041*)
	C LASSO	.617 (*.280*)	**.422** (*.261*)	**.439** (*.184*)	.623 (*.052*)	.826 (*.215*)	**.410** (*.212*)	.501 (*.158*)	.689 (*.053*)
S2	PRIMED	**.654** (*.315*)	.422 (*.254*)	.435 (*.180*)	**.636** (*.057*)	**.909** (*.135*)	.366 (*.153*)	**.500** (*.129*)	**.726** (*.035*)
	CD GL	.589 (*.294*)	.359 (*.224*)	.398 (*.187*)	.615 (*.056*)	.866 (*.177*)	.364 (*.154*)	.477 (*.110*)	.699 (*.042*)
	C LASSO	.617 (*.283*)	**.426** (*.263*)	**.439** (*.184*)	.623 (*.052*)	.829 (*.210*)	**.396** (*.194*)	.497 (*.157*)	.689 (*.052*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

Table 8 Mid-dimensional simulation: DGP 2 ($\gamma = 0$, K=50, n=200, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.871** (*.148*)	**.518** (*.169*)	**.623** (*.115*)	**.684** (*.035*)	**.949** (*.107*)	**.538** (*.166*)	**.665** (*.120*)	**.746** (*.029*)
	CD GL	.686 (*.294*)	.500 (*.273*)	.497 (*.186*)	.643 (*.052*)	.917 (*.112*)	.478 (*.156*)	.609 (*.113*)	.720 (*.033*)
	C LASSO	.829 (*.185*)	.486 (*.204*)	.570 (*.135*)	.674 (*.034*)	.894 (*.147*)	.479 (*.193*)	.592 (*.135*)	.710 (*.033*)
S2	PRIMED	**.874** (*.140*)	**.521** (*.177*)	**.624** (*.119*)	**.684** (*.034*)	**.951** (*.106*)	**.533** (*.156*)	**.663** (*.113*)	**.744** (*.028*)
	CD GL	.686 (*.296*)	.487 (*.268*)	.491 (*.183*)	.643 (*.053*)	.917 (*.116*)	.473 (*.149*)	.607 (*.115*)	.721 (*.032*)
	C LASSO	.834 (*.194*)	.490 (*.204*)	.572 (*.133*)	.675 (*.034*)	.891 (*.143*)	.473 (*.180*)	.590 (*.130*)	.710 (*.032*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

Table 9 High-dimensional simulation: DGP 2 ($\gamma = 0$, K=100, n=200, τ=0.3)

		SNR = 0.5				SNR = 1.0			
		TPR	PPV	F1	AUC	TPR	PPV	F1	AUC
S1	PRIMED	**.700** (*.280*)	**.476** (*.230*)	**.495** (*.168*)	**.647** (*.046*)	**.923** (*.120*)	**.471** (*.183*)	**.595** (*.135*)	**.729** (*.028*)
	CD GL	.531 (*.315*)	.411 (*.254*)	.394 (*.187*)	.611 (*.055*)	.780 (*.212*)	.420 (*.228*)	.481 (*.132*)	.682 (*.042*)
	C LASSO	.620 (*.294*)	.364 (*.213*)	.407 (*.179*)	.629 (*.053*)	.780 (*.242*)	.394 (*.209*)	.479 (*.159*)	.676 (*.050*)
S2	PRIMED	**.694** (*.283*)	**.470** (*.225*)	**.485** (*.165*)	**.647** (*.045*)	**.926** (*.116*)	**.461** (*.170*)	**.591** (*.127*)	**.728** (*.029*)
	CD GL	.514 (*.319*)	.402 (*.261*)	.383 (*.195*)	.609 (*.057*)	.777 (*.218*)	.421 (*.228*)	.481 (*.136*)	.682 (*.043*)
	C LASSO	.626 (*.298*)	.359 (*.214*)	.403 (*.179*)	.629 (*.053*)	.786 (*.242*)	.397 (*.210*)	.482 (*.160*)	.675 (*.050*)

Mean (*SD*) over $B = 50$ replications; best mean in bold.

3.4 Application: rectal cancer pCR prediction

3.4.1 Data and design

We apply PRIMED to a multi-reader MRI study of rectal cancer patients. The dataset comprises 149 patients from three centers who received neoadjuvant chemoradiotherapy. Five radiologists independently evaluated 14 MRI features for each patient: 4 binary features and 10 ordinal features. The outcome is pathologic complete response (pCR).

For each ordinal feature k, the consensus $\bar{X}_k$ is defined as the mean of the five reader scores and the disagreement S_k as the standard deviation. Binary features enter the model as their reader-averaged proportions only, because their disagreement is a deterministic function of the consensus (Appendix A.4).

We split the data by center: Center 1 ($n = 50$, pCR rate 34.0%) serves as the training set, and Centers 2 and 3 ($n = 99$, pCR rate 15.2%) serve as the external validation set.

3.4.2 Dispersion structure

Before fitting models, we examine the consensus–disagreement relationship in this dataset. Figure 6 visualizes the Pearson correlation between $\bar{X}_k$ and S_k for each ordinal feature. Eight of the ten features show correlations whose 95% CI excludes zero, confirming that reader disagreement is systematically related to the consensus level. Pre_DWI shows the strongest relationship ($r = -0.88$, 95% CI: $[-0.92, -0.85]$), whereas Post_cT ($r = -0.09$) and MERCURY ($r = +0.06$) have CIs containing zero.

Inspection of Figure 6 reveals that several features exhibit nonlinear patterns, in particular inverted-U shapes where disagreement peaks at intermediate consensus values and drops near the scale boundaries. The implications of these nonlinear patterns for model specification are explored in the sensitivity analysis (Section 3.4.4).

3.4.3 Model comparison

Table 10 compares the variables selected by PRIMED, CD GL, and C LASSO. PRIMED selects two ordinal features: Fibrosis grading ($\hat{\alpha} = -0.17$, $\hat{\beta} = -1.00$, $\hat{\gamma} = 0.09$) and Pre_DWI ($\hat{\alpha} = -0.46$, $\hat{\beta} = 0.73$, $\hat{\gamma} = -0.14$). CD GL selects three variables: Fibrosis grading, ESGAR, and the binary feature Split scar. Pre_DWI is selected by PRIMED but not by CD GL or C LASSO; conversely, ESGAR and Split scar are selected by both CD GL and C LASSO but dropped by PRIMED. C LASSO, which uses only the consensus channel, selects Fibrosis grading ($\hat{\beta} = -0.66$), ESGAR ($\hat{\beta} = -0.10$), and the binary feature Split scar ($\hat{\delta} = 1.36$) as consensus-only predictors (Table 10).

Table 10 Variable selection comparison on rectal cancer data

		PRIMED		CD GL		C LASSO
	Feature	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\beta}$
Selected	Fibrosis grading	−1.00	0.09	−0.40	−0.10	−0.66
	Pre_DWI	0.73	−0.14			
	ESGAR			−0.55	−0.36	−0.10
Binary	Split scar			$\hat{\delta} = 2.56$		1.36

Only features selected by at least one method are shown. Empty cells indicate non-selection. PRIMED and CD GL estimate $(\hat{\beta}, \hat{\gamma})$ for ordinal features; C LASSO estimates consensus-only $\hat{\beta}$. Binary features enter with a single coefficient $\hat{\delta}$.

Table 10 summarizes the estimated coefficients; the corresponding fitted models are:

$$\begin{aligned}
\text{PRIMED (linear):}\quad \hat{p}_i &= \text{logit}^{-1}\big(\hat{\beta}_0 + \hat{\beta}_B\,\bar{X}_{i,B} + \hat{\gamma}_B\,\tilde{S}_{i,B} + \hat{\beta}_F\,\bar{X}_{i,F} + \hat{\gamma}_F\,\tilde{S}_{i,F}\big),\\
\text{CD GL:}\quad \hat{p}_i &= \text{logit}^{-1}\big(\hat{\beta}_0 + \hat{\beta}_B\,\bar{X}_{i,B} + \hat{\gamma}_B\,S_{i,B} + \hat{\beta}_A\,\bar{X}_{i,A} + \hat{\gamma}_A\,S_{i,A} + \hat{\delta}_C\,\bar{X}_{i,C}\big),\\
\text{C LASSO:}\quad \hat{p}_i &= \text{logit}^{-1}\big(\hat{\beta}_0 + \hat{\beta}_B\,\bar{X}_{i,B} + \hat{\beta}_A\,\bar{X}_{i,A} + \hat{\delta}_C\,\bar{X}_{i,C}\big),
\end{aligned}$$

where $\tilde{S}_{i,k} = S_{i,k} - \hat{\alpha}_{0k} - \hat{\alpha}_k \bar{X}_{i,k}$ is the residual disagreement for feature k under the linear dispersion model. PRIMED uses residual disagreement $\tilde{S}_k$ (after removing the consensus–disagreement dependence), CD GL uses raw disagreement S_k alongside consensus, and C LASSO uses only the consensus channel.

Table 11 reports the training and external validation AUC. Prediction performance is measured by the area under the receiver operating characteristic curve (AUC): the training AUC is computed on Center 1 ($n = 50$) using the fitted probabilities from each method, and the external AUC is computed on Centers 2–3 ($n = 99$) by applying the trained model to the held-out centers without refitting. Training AUCs are similar across methods (0.865–0.871). CD GL achieves the highest external AUC (0.782), followed by C LASSO (0.767) and PRIMED (0.753).

Table 11 Prediction performance on rectal cancer data (AUC)

Method	Variables selected	Training AUC	External AUC
PRIMED	2	0.865	0.753
CD GL	3	0.871	**0.782**
C LASSO	3	0.865	0.767

Training: Center 1 ($n = 50$); External: Centers 2–3 ($n = 99$).

To verify that the consensus–disagreement relationship for the selected features is genuine, we fit separate OLS regressions of S_k on $\bar{X}_k$ for each selected feature. Both show significant dispersion slopes: Pre_DWI ($\hat{\alpha}_{\text{OLS}} = -1.54$, $p < 0.001$) and Fibrosis grading ($\hat{\alpha}_{\text{OLS}} = -0.17$, $p = 0.040$), confirming that the link is real and not an artifact of the joint estimation.

The OLS estimate for Pre_DWI ($\hat{\alpha}_{\text{OLS}} = -1.54$) differs substantially from the PRIMED joint estimate ($\hat{\alpha} = -0.46$ in Table 10). This discrepancy provides direct empirical evidence of the coupling gradient mechanism described in Section 2.5: because $\hat{\gamma}_F \neq 0$, the outcome loss pulls $\hat{\alpha}_F$ away from the OLS solution toward a

value that jointly optimizes both the dispersion and outcome objectives. For Fibrosis grading, where $|\hat{\gamma}|$ is small (0.09), the PRIMED estimate ($\hat{\alpha} = -0.17$) closely matches the OLS value, consistent with the theoretical prediction that the coupling gradient vanishes when $\gamma_k \approx 0$.

Pre_DWI warrants separate comment. PRIMED's penalized estimation selects this feature with a well-defined group norm ($\|(\hat{\beta}, \hat{\gamma})\|_2 = 0.74$), demonstrating that it carries outcome-relevant information through both the consensus and residual disagreement channels. In the unpenalized refit, however, the near-perfect consensus–disagreement correlation ($r = -0.88$) induces collinearity between $\bar{X}_k$ and $\tilde{S}_k$, leading to inflated standard errors for the individual coefficients. This distinction between selection and estimation stability is expected: PRIMED's group penalty regularizes the coefficient estimates during selection, whereas the unpenalized refit exposes the inherent collinearity. The selection itself remains reliable; what is unstable is the decomposition of the effect into β_k and γ_k in the absence of regularization. The selection of Pre_DWI is revisited in the sensitivity analysis below, where a quadratic dispersion model reveals that its apparent disagreement signal is largely an artifact of linear misspecification.

3.4.4 Sensitivity analysis: quadratic dispersion model

To quantify the nonlinear consensus–disagreement patterns, we compare linear and quadratic dispersion models for each ordinal feature. Table 12 reports the results: Fibrosis grading and Post_restriction show the largest gains ($\Delta R^2 > 0.60$), and Pre_DWI achieves $R^2 = 0.992$ under the quadratic model. Given these nonlinear patterns, we refit PRIMED with the quadratic dispersion model (6) on the same training data.

Table 13 compares the results. With a more flexible dispersion model that captures the inverted-U boundary effect, the quadratic PRIMED selects only Fibrosis grading ($\hat{\beta} = -0.63$, $\hat{\gamma} = -0.05$), the feature that was selected by all three methods under the

Table 12 Dispersion model comparison: linear vs quadratic R^2 ($n = 149$)

Feature	R^2 (linear)	R^2 (quadratic)	ΔR^2	$\hat{\alpha}_2$
Fibrosis grading	0.085	0.695	+0.609	−0.311***
Post_restriction	0.076	0.679	+0.604	−0.391***
Pre_cT	0.117	0.412	+0.296	−0.197***
Post_DWI	0.285	0.501	+0.216	−0.315***
Pre_DWI	0.781	0.992	+0.211	−0.750***
ESGAR	0.169	0.358	+0.189	−0.453***
Post_cT	0.009	0.166	+0.158	−0.110***
MERCURY	0.003	0.129	+0.126	−0.140***
Pre_ADC	0.261	0.312	+0.051	+0.757**
Post_ADC	0.135	0.135	+0.000	+0.010

R^2 from OLS regression of S_k on $\bar{X}_k$ (linear) or $\bar{X}_k$ and $\bar{X}_k^2$ (quadratic), computed on all 149 subjects. ***$p < 0.001$, **$p < 0.01$ for the quadratic coefficient $\hat{\alpha}_2$.

linear specification. The fitted model is

$$\text{PRIMED (quadratic):} \quad \hat{p}_i = \text{logit}^{-1}\big(\hat{\beta}_0 + \hat{\beta}_B\, \bar{X}_{i,B} + \hat{\gamma}_B\, \tilde{S}_{i,B}^{\text{quad}}\big),$$

where $\tilde{S}_{i,B}^{\text{quad}} = S_{i,B} - \hat{\alpha}_{0B} - \hat{\alpha}_{1B}\, \bar{X}_{i,B} - \hat{\alpha}_{2B}\, \bar{X}_{i,B}^2$ is the residual disagreement under the quadratic dispersion model. Pre_DWI, which the linear PRIMED selected, is no longer retained: the quadratic dispersion model explains 99.2% of its consensus–disagreement variance (Table 12), leaving negligible residual disagreement with which to predict the outcome.

Table 13 Sensitivity analysis: linear vs quadratic dispersion model

Method	Variables	λ^*	Training AUC	External AUC
PRIMED (linear)	2	0.010	0.865	0.753
PRIMED (quadratic)	1	0.113	0.852	0.766
CD GL	3	—	0.871	0.782
C LASSO	3	—	0.865	0.767

The quadratic PRIMED selects only Fibrosis grading. Despite using a single feature, its external AUC (0.766) matches C LASSO (0.767, 3 variables).

Despite using only one feature, the quadratic PRIMED achieves an external AUC of 0.766, essentially matching C LASSO (0.767 with three variables) and narrowing the gap with CD GL (0.782). The more parsimonious model also avoids the overfitting risk associated with small training samples: the training–external AUC gap is 0.086 for the quadratic PRIMED versus 0.112 for the linear PRIMED. This result illustrates that a better-specified dispersion model can improve variable selection parsimony without sacrificing prediction performance.

3.5 Application: LIDC-IDRI lung nodule radiomics

3.5.1 Data and design

To validate PRIMED in a high-dimensional radiomics setting, we apply it to the Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) dataset [3]. The dataset comprises 695 patients with 1,390 lung nodules, each annotated by 3–4 radiologists. A CT windowing of $[-1000, 400]$ Hounsfield units (HU) was applied to remove soft-tissue artifacts and standardize voxel intensity ranges across scans (see Appendix D for details). For each nodule, $K = 107$ PyRadiomics features [18] were extracted from the individual radiologist segmentations, yielding a multi-reader radiomics matrix with consensus and disagreement summaries computed as described in Section 2.1. The binary outcome is malignancy status (malignant vs. benign), derived from diagnostic information.

Prediction performance was evaluated via nested cross-validation: a stratified 10-fold outer CV assessed generalization, while within each outer training fold, a stratified 5-fold inner CV selected λ (using the same data-driven path and deviance criterion described in Section 2.7). The same nested structure was used for all three methods: PRIMED used 5-fold inner CV over the data-driven λ path, CD GL and C LASSO used the default cross-validation procedures of `grpreg` and `glmnet`, respectively, each with 5 inner folds. This design ensures that no information from the outer test fold

leaks into the λ selection, and that all methods are compared under identical train/test partitions.

Note that the rectal cancer application (Section 3.4) uses a different evaluation strategy: training on a single center and external validation on held-out centers, without outer CV, because the multi-center design provides a natural external validation set.

3.5.2 Prediction and variable selection

Table 14 reports the 10-fold CV AUC and the number of features selected. All three methods achieve comparable prediction performance (AUC 0.898–0.905), with pairwise differences smaller than the within-method standard deviation across folds (SD $\approx$ 0.033–0.035). PRIMED selects the most features on average (65.6), followed by CD GL (30.7) and C LASSO (23.2).

Table 14 LIDC-IDRI lung nodule radiomics: 10-fold CV performance

Method	AUC (SD)	Features selected
PRIMED	0.900 (0.035)	65.6
CD GL	0.905 (0.033)	30.7
C LASSO	0.898 (0.034)	23.2

Mean (SD) over 10 stratified CV folds with nested 5-fold inner CV for λ selection.

3.5.3 Variable selection stability

Beyond point prediction, we assess the consistency of variable selection across CV folds using the Jaccard index [19]: for two sets of selected features A and B, $J(A, B) = |A \cap B|/|A \cup B|$. The Jaccard index is averaged over all $\binom{10}{2} = 45$ pairs of folds. Table 15 reports the results. PRIMED achieves the highest selection stability ($J = 0.798$, SD 0.053), substantially exceeding CD GL (0.688, SD 0.053) and C LASSO (0.560,

SD 0.126). The high Jaccard index indicates that PRIMED's selected feature set is robust to the choice of training fold, a desirable property for clinical deployment where reproducibility is essential.

Table 15 LIDC-IDRI: variable selection stability (Jaccard index)

Method	Jaccard (SD)	Features selected
PRIMED	**0.798** (0.053)	65.6
CD GL	0.688 (0.053)	30.7
C LASSO	0.560 (0.126)	23.2

Jaccard index averaged over all $\binom{10}{2} = 45$ pairs of CV folds.

4 Discussion

PRIMED's advantage stems from three design choices that work in concert.

By modeling the consensus–disagreement relationship via $\tilde{S}_{ik} = S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik}$ and feeding only the residual into the outcome model, PRIMED prevents features with large dispersion effects but no outcome relevance (noise variables) from being selected. CD GL, which enters raw S_k, conflates dispersion structure with outcome signal; C LASSO discards the disagreement channel entirely and collapses when subject-specific noise is large ($\tau \geq 0.6$).

Because α_k is unpenalized, the consensus–disagreement relationship is estimated without penalty-induced bias regardless of whether feature k is selected. Variable selection is driven solely by $\|(\beta_k, \gamma_k)\|_2$, ensuring that inactive features with nontrivial α_k are absorbed into the dispersion model rather than triggering false positives. This mechanism is most valuable in high dimensions: at $K = 100$, PRIMED consistently outperforms CD GL in F1, driven primarily by higher PPV.

Because $\mathcal{L}_{\text{out}}$ depends on α_k through $\tilde{S}_{ik}$, the dispersion parameters receive outcome-driven gradient updates. The rectal cancer data provide direct evidence: for

Pre_DWI ($\hat{\gamma} \neq 0$), the PRIMED estimate $\hat{\alpha} = -0.46$ departs substantially from the OLS estimate $\hat{\alpha}_{\text{OLS}} = -1.54$, whereas for Fibrosis ($\hat{\gamma} \approx 0$) the two coincide.

Beyond point accuracy, PRIMED produces the most reproducible feature sets. In simulations, the Jaccard index is highest at all dimensionalities (0.507/0.352/0.228 for $K = 20/50/100$), with the gap over competitors widening in higher dimensions. In the LIDC-IDRI application ($K = 107$), PRIMED achieves Jaccard 0.798 versus 0.688 (CD GL) and 0.560 (C LASSO), indicating that approximately 80% of selected features are shared across CV folds.

The two applications span different imaging modalities (MRI vs. CT), clinical tasks (rectal cancer pCR vs. lung nodule malignancy), feature types (ordinal scores vs. continuous radiomics), and dimensionalities ($K = 14$ vs. $K = 107$). PRIMED's consistent advantages across these settings suggest that the framework generalizes beyond a single clinical context.

The joint objective is non-convex, though empirical path analysis shows strictly monotone regularization paths on the LIDC-IDRI data (Section 2.5). The rectal cancer training sample ($n = 50$) limits the external AUC of PRIMED (0.753) relative to CD GL (0.782); simulation results at larger sample sizes consistently favor PRIMED. The quadratic dispersion extension (Section 3.4.4) improves model specification for features with inverted-U patterns, but more flexible specifications (splines, higher-order polynomials) remain to be explored. The proximal gradient algorithm is slower than off-the-shelf group LASSO solvers. Future work could extend PRIMED to survival or multi-class outcomes.

5 Conclusions

We proposed PRIMED, a joint loss framework for multi-reader imaging data that integrates consensus, disagreement, and their structural relationship into a single

penalized objective. The residual disagreement formulation removes the consensus–disagreement dependence before entering disagreement into the outcome model, while selective penalization leaves dispersion parameters unpenalized and drives variable selection solely through the outcome coefficients. Simulations demonstrate consistent superiority in variable selection accuracy and stability across a range of dimensions, noise levels, and signal strengths. Two applications validate the framework: in rectal cancer MRI ($K = 14$), PRIMED identifies clinically meaningful signals missed by competing methods; in LIDC-IDRI lung nodule radiomics ($K = 107$), it achieves comparable prediction with substantially higher selection stability (Jaccard 0.798 vs. 0.688 and 0.560).

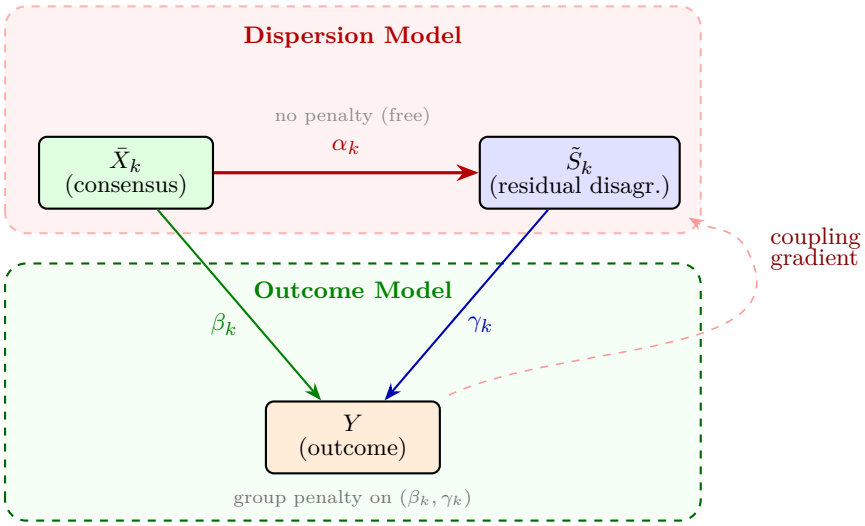

Fig. 2 Path diagram of the PRIMED model for feature k. **Dispersion model (top):** A linear regression links consensus $\bar{X}_k$ to disagreement via slope α_k, estimated without penalty. The residual disagreement $\tilde{S}_k$ captures the component of reader disagreement not explained by the consensus level. **Outcome model (bottom):** A logistic regression predicts Y using consensus (via β_k) and residual disagreement (via γ_k), with a group penalty on (β_k, γ_k). Feature k is selected when $(\hat{\beta}_k, \hat{\gamma}_k) \neq (0, 0)$. The dashed arrow indicates the coupling gradient: because $\tilde{S}_k$ depends on α_k, the outcome loss feeds gradient updates to the dispersion parameters, achieving joint estimation.

List of abbreviations. pCR: pathologic complete response; CRT: chemoradiation therapy; MRI: magnetic resonance imaging; CD: consensus–disagreement; PRIMED:

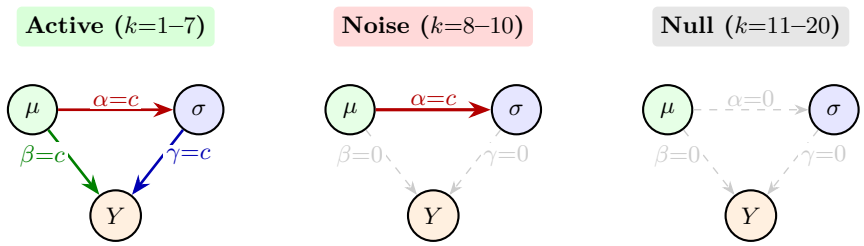

Fig. 3 Variable structure in the noise simulation. Active variables (S2) have all three paths nonzero. Noise variables have $\alpha \neq 0$ (reader disagreement present) but $\beta = \gamma = 0$ (no outcome relevance). Null variables have no effect on any path. In S1, noise variables are identical to null variables ($\alpha_{8-10} = 0$).

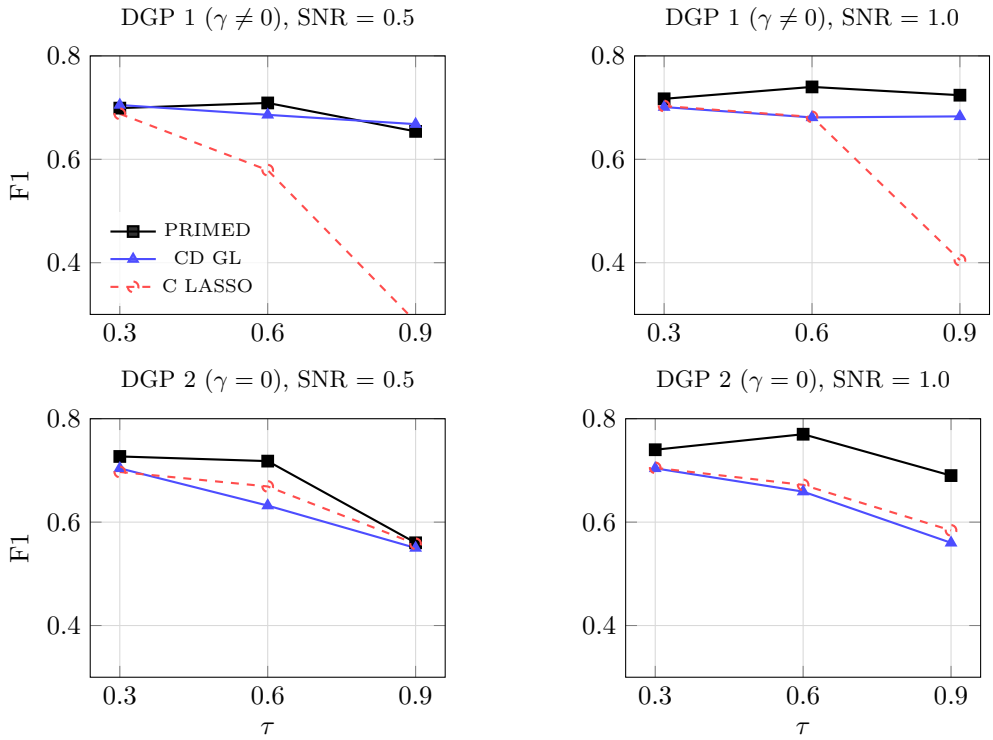

Fig. 4 F1 score as a function of subject-specific noise τ. Results shown for S2 (noise present) under DGP 1 (top) and DGP 2 (bottom). Under DGP 1, C LASSO collapses at high τ because it ignores the disagreement channel that carries outcome signal. Under DGP 2, C LASSO remains stable since $\gamma = 0$, but PRIMED still achieves the highest F1 through superior PPV.

Penalized Regularization for Integrated Multi-path Estimation with Dispersion; GL: group LASSO; C LASSO: consensus LASSO; OLS: ordinary least squares; DGP: data generating process; SNR: signal-to-noise ratio; TPR: true positive rate; PPV: positive predictive value; AUC: area under the receiver operating characteristic curve;

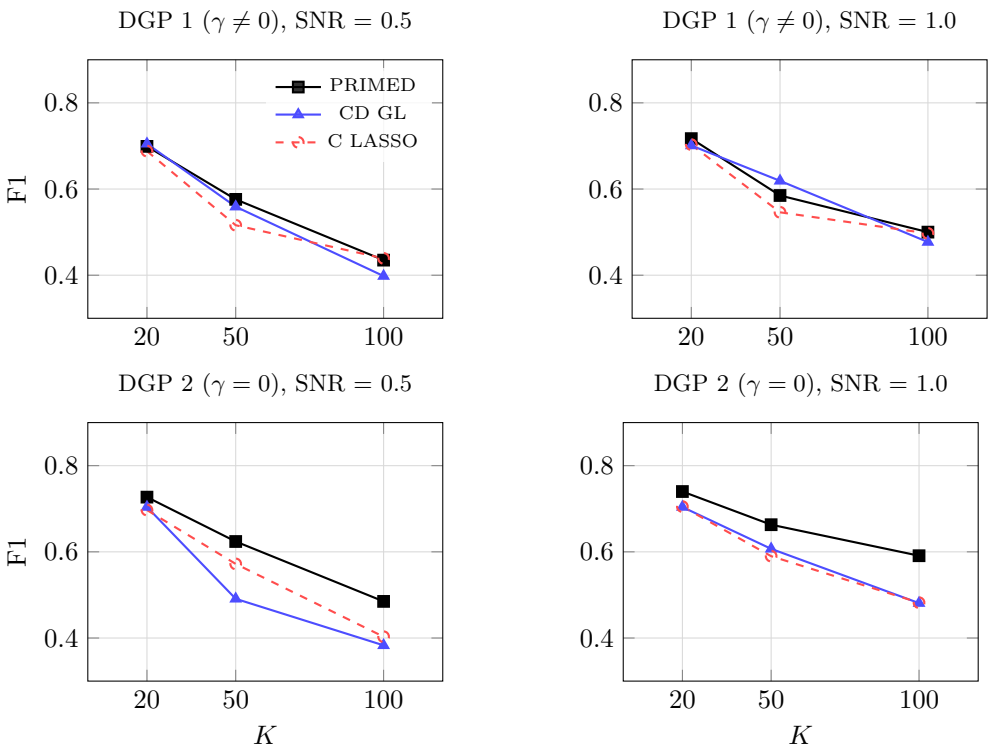

Fig. 5 F1 score as a function of dimensionality K. Results shown for S2 (noise present) under DGP 1 (top) and DGP 2 (bottom). Under DGP 1, PRIMED's AUC advantage is consistent across K. Under DGP 2, PRIMED maintains a consistent advantage (F1 gap $> .100$ at $K = 100$), driven by its ability to maintain higher PPV when many inactive features have nontrivial dispersion structure.

DWI: diffusion-weighted imaging; ADC: apparent diffusion coefficient; ESGAR: European Society of Gastrointestinal and Abdominal Radiology; ICC: intraclass correlation coefficient; SD: standard deviation; LIDC-IDRI: Lung Image Database Consortium and Image Database Resource Initiative; HU: Hounsfield units; CT: computed tomography; CV: cross-validation;

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board of Severance Hospital, Yonsei University College of Medicine (IRB approval number to be inserted).

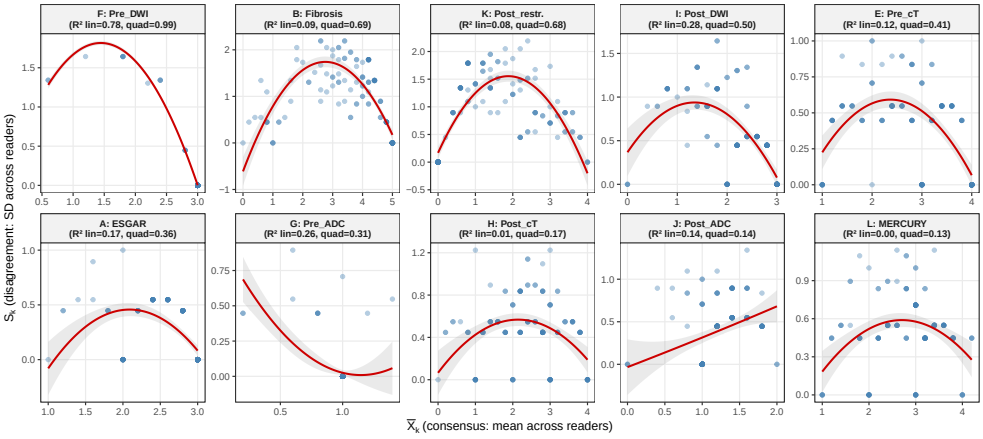

Fig. 6 Consensus–disagreement scatter plots for each ordinal feature. The horizontal axis shows the consensus $\bar{X}_k$ (mean across five readers, $n = 149$) and the vertical axis shows the disagreement S_k (standard deviation across readers). Red curves indicate the fitted quadratic regression $S_k = \alpha_{0k} + \alpha_{1k}\bar{X}_k + \alpha_{2k}\bar{X}_k^2 + \varepsilon_k$; shaded bands show 95% confidence intervals. Panel titles report R^2 for both the linear and quadratic models. Features are ordered by quadratic R^2; Pre_DWI ($R^2 = 0.99$), Fibrosis ($R^2 = 0.69$), and Post_restriction ($R^2 = 0.68$) show pronounced inverted-U patterns that the linear model fails to capture.

Informed consent was obtained from all participants. The LIDC-IDRI dataset is publicly available and does not require separate ethics approval.

Consent for publication

Not applicable.

Availability of data and materials

The LIDC-IDRI dataset analysed during the current study is publicly available in The Cancer Imaging Archive (TCIA) repository [20]. The rectal cancer MRI dataset used during the current study is available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

Funding

Not applicable.

Authors' contributions

YK conceived the study, developed the methodology, implemented the algorithms, performed all analyses, and drafted the manuscript. KH supervised the study and critically revised the manuscript. All authors read and approved the final manuscript.

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Not applicable.

Appendix A Consensus–disagreement measures for multi-reader data

PRIMED requires two subject-level summaries per feature: a consensus measure and a disagreement (dispersion) measure. In Section 2 we use the sample mean and sample standard deviation, which are the natural choices when features are continuous or ordinal with five or more categories. This appendix catalogues alternatives for different measurement scales, so that PRIMED can be applied to a broader range of multi-reader data. The two applications in this paper illustrate different ends of this spectrum: the rectal cancer study (Section 3.4) uses ordinal MRI features scored on 3–6 point scales ($K = 10$, $m = 5$ readers), while the LIDC-IDRI study (Section 3.5) uses continuous radiomics features extracted from 3D segmentation masks ($K = 107$, $m = 3$–4 readers).

Throughout, R readers evaluate a subject on a single feature, producing scores $X_1, \ldots, X_R$.

A.1 Guiding principles

Subject-level measures are required.

Global agreement indices such as ICC [4] and Fleiss' κ [5] produce a single number for the entire dataset. PRIMED, by contrast, needs one consensus value and one disagreement value *per subject per feature*, so that each patient's degree of reader disagreement can enter the model as a predictor.

When disagreement is informative.

The dispersion model $S_k = \alpha_{0k} + \alpha_k \bar{X}_k + \varepsilon_k$ is most useful when $\alpha_k \neq 0$, i.e., when disagreement depends systematically on the consensus level. Common patterns include negative linear (disagreement decreases near the scale boundaries), positive linear, and inverted-U-shaped (disagreement peaks at intermediate scores). When nonlinear patterns are present, the quadratic extension $S_k = \alpha_{0k} + \alpha_{1k}\bar{X}_k + \alpha_{2k}\bar{X}_k^2 + \varepsilon_k$ (Section 2.8) can substantially improve the explained variance; e.g., from $R^2 = 0.085$ to 0.695 for Fibrosis in the rectal cancer data (Table 12). In the LIDC-IDRI radiomics data, 77 of 107 features show higher quadratic than linear R^2 (Section 3.5). If S_k is approximately constant across subjects, the dispersion channel adds little information.

A.2 Continuous features

For continuous features, the arithmetic mean and standard deviation are the default consensus and disagreement measures used in PRIMED (Section 2). This is the setting used in the LIDC-IDRI radiomics application (Section 3.5), where each feature is a continuous quantity (e.g., volume in mm^3, entropy in nats) computed independently from each reader's segmentation mask. When features span very different scales, standardization prior to model fitting is recommended; in our implementation, both $\bar{X}_k$ and S_k are centered and scaled to unit variance within the training set. Table A1 lists alternatives.

Table A1 Consensus and dispersion measures for continuous features

Type	Measure	Formula	Properties
Consensus	Arithmetic mean	$\bar{X} = R^{-1} \sum_{r=1}^{R} X_r$	Default in PRIMED
	Median	$\mathrm{Med}(X_1, \ldots, X_R)$	Robust to outliers
	Trimmed mean	Mean after removing top/bottom $k\%$	Compromise
Dispersion	Standard deviation (SD)	$\left[\frac{1}{R-1} \sum_{r=1}^{R} (X_r - \bar{X})^2\right]^{1/2}$	Default in PRIMED
	Range	$\max_r X_r - \min_r X_r$	Simple
	Mean absolute deviation	$R^{-1} \sum_{r=1}^{R} \lvert X_r - \bar{X} \rvert$	Robust to outliers
	Coefficient of variation	$\mathrm{SD}/\bar{X}$	Scale-invariant

A.3 Ordinal features

For ordinal scales with five or more categories, treating the data as approximately continuous and using the mean and SD is well supported [21, 22]. In the rectal cancer application (Section 3.4), all 10 ordinal MRI features (scored on 3–6 point scales by five radiologists) are summarized using the mean and SD, following this continuous approximation. For shorter scales (3–4 categories), ordinal-specific dispersion measures may be preferred as alternatives.

Within-group agreement (r_{WG}).

The r_{WG} index [23] compares the observed variance S_X^2 to the expected variance σ_E^2 under random responding: $r_{WG} = 1 - S_X^2/\sigma_E^2$. Under a uniform null on A response options, $\sigma_E^2 = (A^2 - 1)/12$.

Average deviation (AD_M).

The average deviation around the mean [24], $AD_M = R^{-1} \sum_{r=1}^{R} |X_r - \bar{X}|$, is more robust to outliers than SD. Agreement is judged adequate when $AD_M < c/6$, where c is the number of response options.

Leik's ordinal dispersion (D).

Leik's D [25] uses the cumulative frequency distribution: $D = 0$ indicates perfect consensus and $D = 1$ indicates maximum dispersion.

Ordinal variation ($1 - l^2$).

Blair and Lacy's ordinal variation [26] measures how far the cumulative distribution lies from a maximum-dispersion configuration, ranging from 0 (perfect consensus) to 1 (maximum dispersion).

Table A2 summarizes the available measures by scale length. Both applications in this paper use the mean and SD for all ordinal features.

Table A2 Available consensus and dispersion measures for ordinal features

Scale	Consensus	Dispersion	Rationale
5+ categories	Mean	SD, r_{WG}, AD_M	Treat as continuous
3–4 categories	Mean or Median	SD, Leik's D, $1 - l^2$	Mean/SD used in this paper; ordinal-specific alternatives

A.4 Binary features

For binary features ($X_r \in \{0, 1\}$), the standard deviation $\sqrt{p(1-p)}$ is a deterministic function of the consensus proportion $p = \bar{X}$. Including both as predictors would introduce perfect multicollinearity. PRIMED therefore uses *only the consensus mean* for binary features, applying a LASSO penalty rather than the group penalty used for ordinal features. This is the approach taken in the rectal cancer application (Section 3.4), where four binary MRI features enter the model as reader-averaged proportions. The LIDC-IDRI application does not involve binary features, as all 107 radiomics features are continuous.

A.5 Nominal features

For nominal features with $C \geq 3$ unordered categories, let p_c denote the proportion of readers selecting category c. Unlike binary features, nominal features admit nontrivial dispersion measures. Table A3 lists the most common options.

Table A3 Consensus and dispersion measures for nominal features ($C \geq 3$ categories)

Measure	Formula	Range
Mode (consensus)	$\arg\max_c p_c$	—
Agreement rate	$\max_c p_c$	$[1/C,\ 1]$
Simpson's diversity	$1 - \sum_{c=1}^{C} p_c^2$	$[0,\ 1 - 1/C]$
Shannon entropy	$-\sum_{c=1}^{C} p_c \log_2 p_c$	$[0,\ \log_2 C]$

A.6 Summary

Table A4 summarizes how PRIMED handles each measurement scale.

Table A4 Summary of consensus and dispersion measures by measurement scale

Scale	Consensus	Dispersion	PRIMED penalty	Application
Continuous	Mean	SD	Group (β_k, γ_k)	LIDC-IDRI
Ordinal	Mean	SD	Group (β_k, γ_k)	Rectal
Binary	Mean $(= p)$	Not applicable	LASSO (δ_k)	Rectal
Nominal ($\geq$3 cat.)	Mode	Entropy or Simpson's D	Group (β_k, γ_k)	—

Appendix B Data-driven regularization path

The regularization parameter λ controls the sparsity of the PRIMED solution. We construct a data-driven path following the approach of Friedman et al. [17].

B.1 Computing $\lambda_{\max}$

At the null model ($\beta_k = \gamma_k = 0$ for all k), the predicted probability is $\hat{p}_i = \bar{Y}$ for all i, and the residual disagreement reduces to $\tilde{S}_{ik} = S_{ik} - \bar{S}_k$. The gradient of the outcome loss with respect to (β_k, γ_k) is

$$g_k = \Big(\frac{1}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (\bar{Y} - Y_i)\bar{X}_{ik}, \;\; \frac{1}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (\bar{Y} - Y_i)(S_{ik} - \bar{S}_k) \Big).$$

The proximal step sets $(\beta_k, \gamma_k) = (0, 0)$ whenever $\|g_k\|_2 \leq \eta\lambda\sqrt{2}$. Therefore, all features are deselected when

$$\lambda \geq \lambda_{\max} \;=\; \frac{1}{\sqrt{2}} \max_{k=1,\ldots,K} \|g_k\|_2.$$

B.2 Log-spaced grid

Given $\lambda_{\max}$, we construct a grid of L values log-spaced on $[\lambda_{\max} \cdot \epsilon,\ \lambda_{\max}]$, where $\epsilon = 0.01$ (simulations, $K \leq 100$) or $\epsilon = 0.001$ (LIDC-IDRI application, $K = 107$, where the larger feature space benefits from finer resolution at small λ):

$$\lambda_j = \lambda_{\max} \cdot \exp\Big(\frac{j-1}{L-1} \log \epsilon \Big), \quad j = 1, \ldots, L.$$

This ensures dense coverage near $\lambda_{\max}$ (where the transition from null to non-null models occurs) and sparser coverage at small λ values.

Appendix C Rectal cancer MRI feature descriptions

Table C5 lists all $K = 14$ MRI features used in the rectal cancer application (Section 3.4). Ten ordinal features are scored on 3–6 point scales by five radiologists; their consensus and disagreement are computed as the sample mean and standard

deviation, respectively. Four binary features are scored as present or absent; their consensus enters the model as the reader-averaged proportion (Appendix A.4).

Table C5 MRI features in the rectal cancer application

Feature	Scale	Categories
Ordinal features (consensus = mean, disagreement = SD)		
Fibrosis grading	Normalized wall to viable tumor	6
Pre_cT	T stage (partial to T3c+/T4)	4
Pre_DWI	DWI signal (not seen / low / iso / high)	4
Pre_ADC	ADC signal (not seen / low / iso / high)	4
Post_cT	T stage (normalized to T3c+)	5
Post_DWI	DWI signal (not seen / low / iso / high)	4
Post_ADC	ADC signal (not seen / low / iso / high)	4
Post_restriction	Restriction (none to diffuse)	5
MERCURY	Tumor regression grade (1–5)	5
ESGAR	Normal / fibrosis only / residual tumor	3
Binary features (consensus = proportion, no disagreement channel)		
Split scar	Absent / present	2
Ulcer	Absent / present	2
Pre_EMVI	Absent / present	2
Normalized wall	Absent / present	2

DWI: diffusion-weighted imaging; ADC: apparent diffusion coefficient; EMVI: extramural vascular invasion; ESGAR: European Society of Gastrointestinal and Abdominal Radiology; MERCURY: Magnetic Resonance Imaging and Rectal Cancer European Equivalence Study.

Appendix D LIDC-IDRI preprocessing details

D.1 CT windowing

CT images contain Hounsfield unit (HU) values spanning a wide range: air appears at approximately −1000 HU, soft tissue at −100 to +100 HU, and cortical bone above +400 HU. Because lung nodules are soft-tissue structures, voxels near the mask boundary may include air (lung parenchyma) or bone (ribs, vertebrae) that are not part of the nodule itself. When different readers draw slightly different contours, these boundary voxels are included in some masks but not others, inflating the measured reader disagreement with artifact rather than genuine diagnostic uncertainty.

We apply CT windowing by clipping all voxel values to $[-1000, 400]$ HU before feature extraction:

$$v_i \leftarrow \max(-1000, \min(400, v_i)).$$

This removes extreme values from air and bone while preserving the full dynamic range of soft tissue.

D.2 PyRadiomics feature classes

The 107 radiomics features extracted per reader per nodule comprise seven classes from the PyRadiomics library [18]: Shape (14 features: volume, surface area, sphericity, elongation, etc.), First-order (18: mean, variance, entropy, skewness, kurtosis, energy, etc.), GLCM (24: contrast, correlation, homogeneity, energy, etc.), GLRLM (16: short/long run emphasis, gray level non-uniformity, etc.), GLSZM (16: small/large area emphasis, zone entropy, etc.), NGTDM (5: coarseness, busyness, complexity, strength, contrast), and GLDM (14: dependence variance, entropy, etc.). All features are extracted with bin width 25, $1 \times 1 \times 1$ mm isotropic resampling, and B-spline interpolation.

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Algorithm 1 PRIMED: Proximal Gradient Descent with Selective Penalization

Require: $\bar{X} \in \mathbb{R}^{n\times K}$, $S \in \mathbb{R}^{n\times K}$, $Y \in \{0,1\}^n$, $\lambda > 0$, step size η, tolerance ϵ, max iterations $T_{\max}$

Ensure: $\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}$

Initialize: $\alpha_k \leftarrow 0$, $\alpha_{0k} \leftarrow \overline{S_k}$, $\beta_k \leftarrow 0$, $\gamma_k \leftarrow 0$, $\beta_0 \leftarrow \mathrm{logit}(\bar{Y})$

Compute null losses: $\mathcal{L}^0_{\text{disp}} \leftarrow \frac{1}{2}\sum_k \widehat{\mathrm{Var}}(S_k)$, $\mathcal{L}^0_{\text{out}} \leftarrow -[\bar{Y}\log\bar{Y} + (1-\bar{Y})\log(1-\bar{Y})]$

for $t = 1, 2, \ldots, T_{\max}$ **do**

  % Dispersion model gradients (normalized by $\mathcal{L}^0_{\text{disp}}$)

  for $k = 1, \ldots, K$ **do**

    $r_{ik} \leftarrow S_{ik} - \alpha_{0k} - \alpha_k \bar{X}_{ik}$

    $\alpha_{0k} \leftarrow \alpha_{0k} + \eta \cdot \frac{1}{n\,\mathcal{L}^0_{\text{disp}}} \sum_i r_{ik}$

    $\alpha_k \leftarrow \alpha_k + \eta \cdot \frac{1}{n\,\mathcal{L}^0_{\text{disp}}} \sum_i r_{ik}\bar{X}_{ik}$ *% dispersion gradient*

    $\alpha_k \leftarrow \alpha_k - \eta \cdot \frac{\gamma_k}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (p_i - Y_i)\bar{X}_{ik}$ *% coupling gradient*

  end for

  % Outcome model gradients (normalized by $\mathcal{L}^0_{\text{out}}$)

  for $i = 1, \ldots, n$ **do**

    $p_i \leftarrow \sigma(\beta_0 + \sum_k \beta_k \bar{X}_{ik} + \gamma_k r_{ik})$ *% $r_{ik} = \tilde{S}_{ik}$*

  end for

  $\beta_0 \leftarrow \beta_0 - \eta \cdot \frac{1}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (p_i - Y_i)$

  for $k = 1, \ldots, K$ **do**

    $\tilde{\beta}_k \leftarrow \beta_k - \eta \cdot \frac{1}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (p_i - Y_i)\bar{X}_{ik}$

    $\tilde{\gamma}_k \leftarrow \gamma_k - \eta \cdot \frac{1}{n\,\mathcal{L}^0_{\text{out}}} \sum_i (p_i - Y_i) r_{ik}$

  end for

  % Proximal step: group soft-thresholding on (β_k, γ_k)

  for $k = 1, \ldots, K$ **do**

    if $\|(\tilde{\beta}_k, \tilde{\gamma}_k)\|_2 > \eta\lambda$ **then**

      $(\beta_k, \gamma_k) \leftarrow \big(1 - \frac{\eta\lambda}{\|(\tilde{\beta}_k, \tilde{\gamma}_k)\|_2}\big)(\tilde{\beta}_k, \tilde{\gamma}_k)$

    else

      $(\beta_k, \gamma_k) \leftarrow (0, 0)$

    end if

  end for

  % Convergence check

  if relative change in parameters $< \epsilon$ **then**

    break

  end if

end for

Selection: feature k is selected if $\hat{\beta}_k \neq 0$ or $\hat{\gamma}_k \neq 0$